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Name:

Teacher:.....



Pymble Ladies' College

**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION
2019**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using pencil for Questions 1-10.
- Write using black or blue pen for Questions 11-14. Black pen is preferred.
- Erasable pens are **not** to be used.
- Board approved calculators may be used.
- A reference sheet is provided.
- In Questions 11-14, show relevant mathematical reasoning and/or calculations.

Total Marks – 70

Section I (Pages 1 – 4)

10 marks

- Attempt all Questions 1-10 on the multiple choice answer sheet provided.
- Allow about 15 minutes for this section

Section II (Pages 5 – 12)

60 marks

- Attempt Questions 11-14
- Allow about 1 hour 45 minutes for this section

Mark	/70
Highest Mark	/70
Rank	

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Section I

10 marks

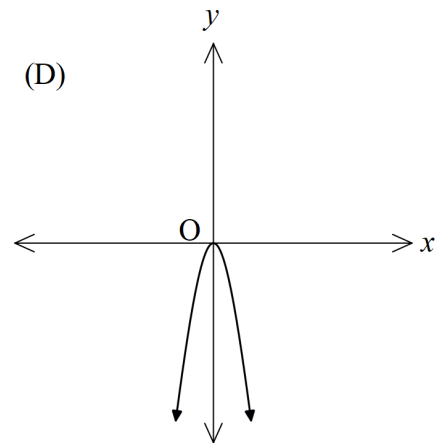
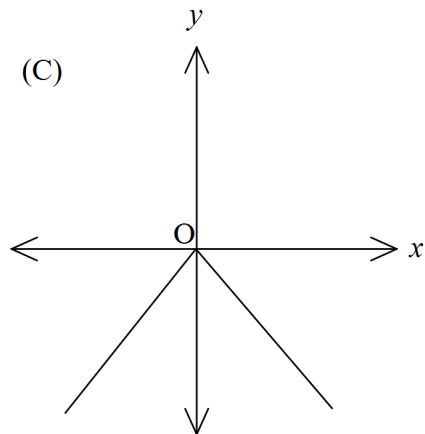
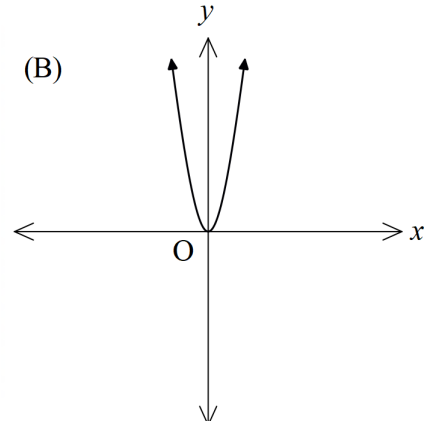
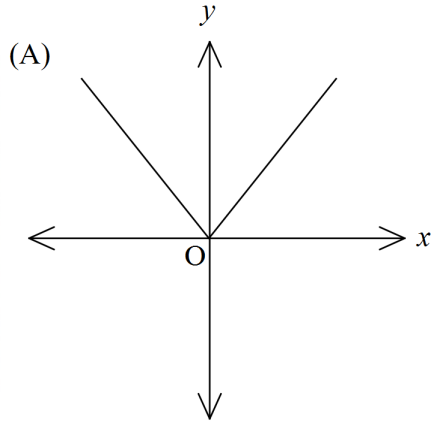
Attempt Questions 1-10

Allow about 15 minutes for this section.

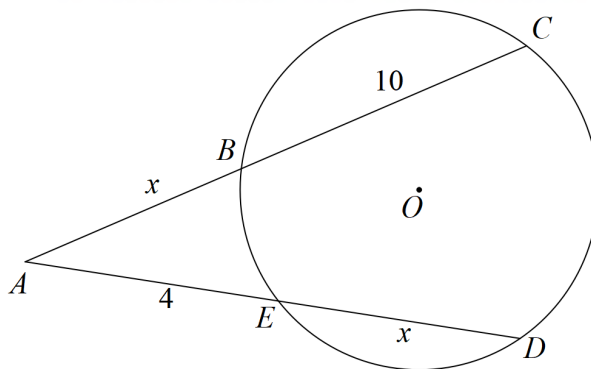
Use the multiple choice answer sheet for Questions 1-10.

- 1 The point A has coordinates $(2, -3)$ and the point B has coordinates $(-4, 5)$.
Find the coordinates of the point which divides AB internally in the ratio 4:3.
- (A) $\left(-\frac{10}{7}, \frac{11}{7}\right)$
- (B) $\left(-\frac{4}{7}, \frac{3}{7}\right)$
- (C) $(-22, 9)$
- (D) $\left(-\frac{25}{7}, \frac{26}{7}\right)$
- 2 What is the remainder when $x^3 - 10x$ is divided by $x + 5$?
- (A) -75
- (B) 75
- (C) $x^2 - 2x$
- (D) $x^2 - 5x + 15$
- 3 What is the cartesian equation of the tangent to the parabola $x = t - 3, y = t^2 + 2$ at $t = -3$?
- (A) $6x - y - 25 = 0$
- (B) $6x - y - 36 = 0$
- (C) $6x + y + 25 = 0$
- (D) $6x + y + 36 = 0$

- 4 What is the best representation of the graph $y = x \tan^{-1} x$?



5

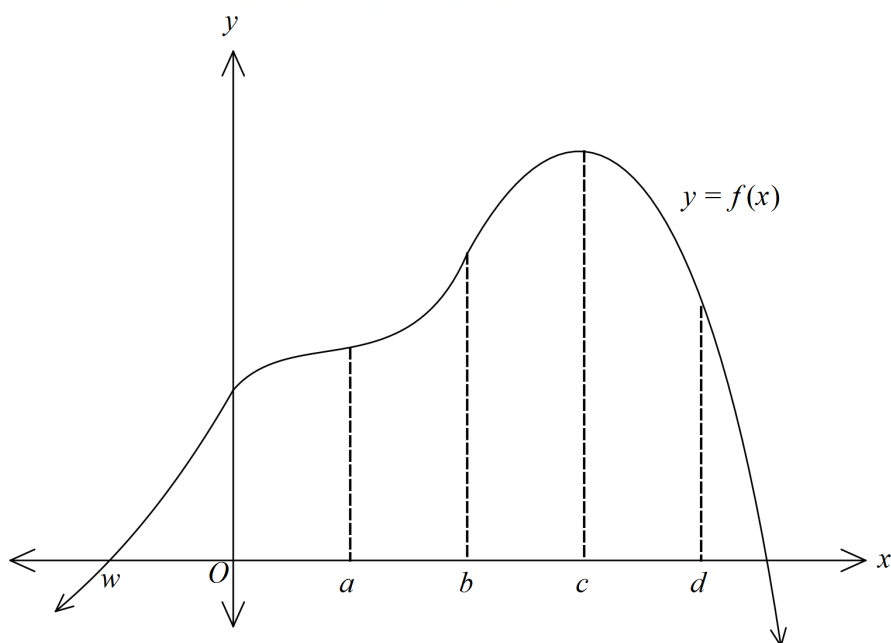


Not to Scale

Two secants from the point A cut the circle centre O at BC and ED .
Given $AB = DE = x$, $BC = 10$ and $EA = 4$, which of the following is the value of x ?

- (A) $x = 6$
- (B) $x = 8$
- (C) $x = 2$
- (D) $x = 5$

- 6 The diagram shows the graph of $y = f(x)$. $f(x) = 0$ has a solution at $x = w$.



Newton's method can be used to give an approximation close to the solution $x = w$. Which initial approximation, x_1 will give the second approximation that is closest to the solution $x = w$?

- (A) $x_1 = a$
(B) $x_1 = b$
(C) $x_1 = c$
(D) $x_1 = d$
- 7 A particle moves in a straight line. Its position at any time t is given by $x = 4 \sin 2t + 3 \cos 2t$. What is the acceleration in terms of x ?
- (A) $\ddot{x} = -4x$
(B) $\ddot{x} = -8x$
(C) $\ddot{x} = -16x^2$
(D) $\ddot{x} = -16 \sin 2x - 12 \cos 2x$

8 What is the general solution to the equation $2\sin^2 \theta - 3\sin \theta - 2 = 0$?

(A) $\theta = n\pi + (-1)^n \frac{\pi}{6}$

(B) $\theta = n\pi + (-1)^n \frac{5\pi}{6}$

(C) $\theta = n\pi + (-1)^n \frac{7\pi}{6}$

(D) $\theta = n\pi + (-1)^n \frac{11\pi}{6}$

9 What is the domain of $f(x) = \sin^{-1}(4x)$?

(A) $-\pi \leq x \leq \pi$

(B) $-4 \leq x \leq 4$

(C) $-\frac{\pi}{8} \leq x \leq \frac{\pi}{8}$

(D) $-\frac{1}{4} \leq x \leq \frac{1}{4}$

10 What is the value of k such that $\int_2^k \frac{dx}{\sqrt{16-x^2}} = \frac{\pi}{6}$?

(A) $\frac{2\pi}{3}$

(B) $2\sqrt{3}$

(C) $\frac{4\pi}{3}$

(D) $\frac{\sqrt{3}}{2}$

Section II

60 marks

Attempt Questions 11-14

Answer each question in a SEPARATE writing booklet. Extra booklets are available.
Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a Separate Booklet.

Marks

(a) Solve $x - \frac{2}{x} \leq 1$.

2

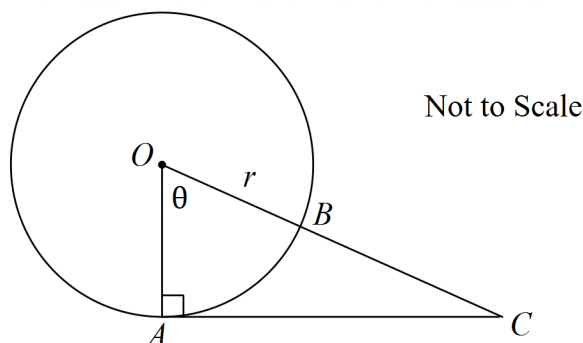
(b) Evaluate $\int_{-1}^0 x\sqrt{1+x} \, dx$ using the substitution $u = 1+x$.

3

(c) Find the exact value of $\tan\left(\frac{5\pi}{12}\right)$, leaving your answer in simplified exact form.

3

- (d) The diagram shows points A and B lying on a circle with centre O and radius r . The arc AB subtends an angle θ at O . The line OB produced and the tangent at A intersect at C .



- (i) It is known that AC is twice the length of the arc AB .
Show that $\tan \theta = 2\theta$

2

- (ii) One solution to the equation $\tan \theta = 2\theta$ is approximately $\theta = 1.2$.
Use one application of Newton's method to find a second approximation.
Give your answer correct to two decimal places.

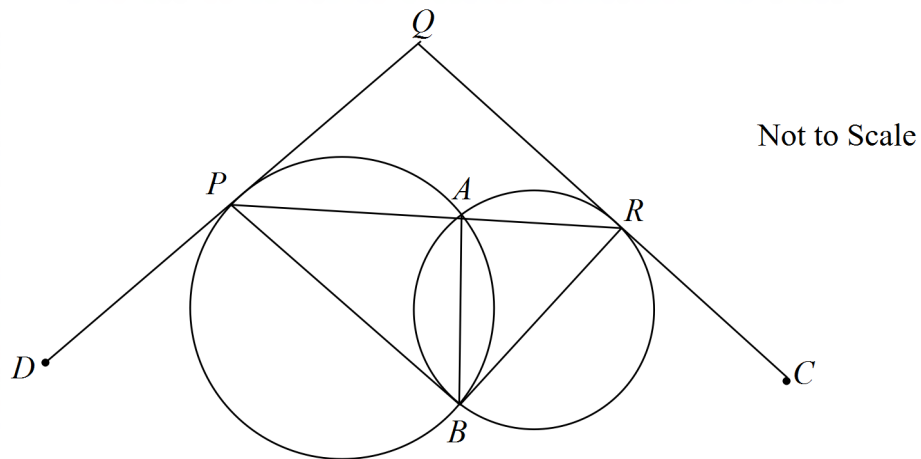
2

Question 11 continues on page 6

Question 11 (continued)

- (e) Two circles intersect at A and B . PAR is a straight line where P is a point on the first circle and R is a point on the second circle. The tangent at P and the tangent at R meet at Q . Prove that $BPQR$ is a cyclic quadrilateral.

3



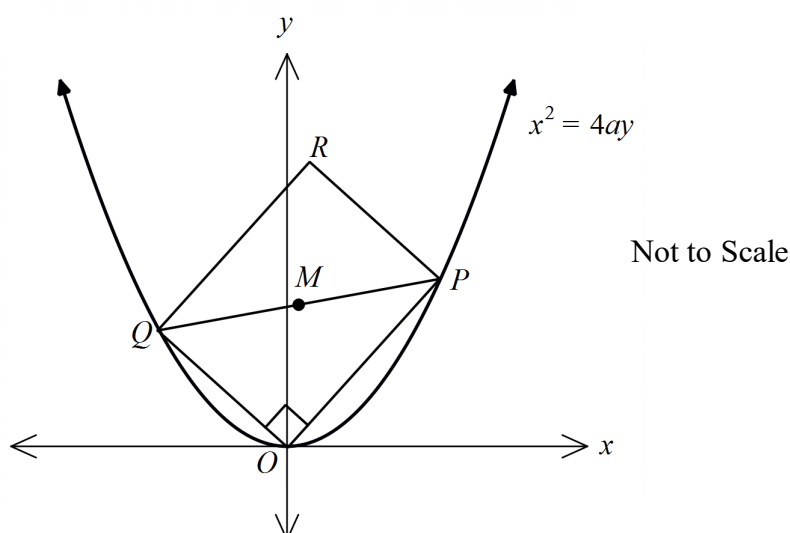
End of Question 11

- (a) Use mathematical induction to prove

3

$$\left(1 - \frac{2}{2 \times 3}\right) \left(1 - \frac{2}{3 \times 4}\right) \left(1 - \frac{2}{4 \times 5}\right) \dots \left(1 - \frac{2}{n(n+1)}\right) = \left(\frac{n+2}{3n}\right) \text{ for all integers } n \geq 2.$$

- (b)



$P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points which move on the parabola $x^2 = 4ay$ such that $\angle POQ = 90^\circ$, where $O(0,0)$ is the origin. $M\left(a(p+q), \frac{1}{2}a(p^2+q^2)\right)$ is the midpoint of PQ . R is the point such that $OPRQ$ is a rectangle.

- (i) Show that $pq = -4$.

1

- (ii) Show that R has coordinates $(2a(p+q), a(p^2+q^2))$.

1

- (iii) Find the equation of the locus of R .

2

Question 12 continues on page 8

Question 12 (continued)

- (c) A sphere is being heated so that its surface area is increasing at a rate of $1.5 \text{ mm}^2/\text{s}$.

Using the formulae for the surface area and volume of a sphere as $A = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$ respectively:

(i) Show that $\frac{dr}{dt} = \frac{3}{16\pi r}$. 2

- (ii) Hence, find the rate at which the volume is increasing when the radius is 60 mm. 2

- (d) A particle is moving in a straight line with $v^2 = 36 - 4x^2$.

- (i) Prove that the motion is simple harmonic. 1

- (ii) Find the period and amplitude of the motion. 2

- (iii) If the particle is initially at the origin, write an expression for its displacement in terms of t . 1

End of Question 12

- (a) A particle moves in a straight line so that at time t seconds, it has acceleration $\alpha \text{ ms}^{-2}$, velocity $v \text{ ms}^{-1}$ and position x metres relative to a fixed point on the line. The velocity and position of the particle at any time t seconds are related by $v = -x^2$. Initially $x = 1$.

- (i) Find the initial acceleration of the particle. 2
- (ii) Express x in terms of t . 2

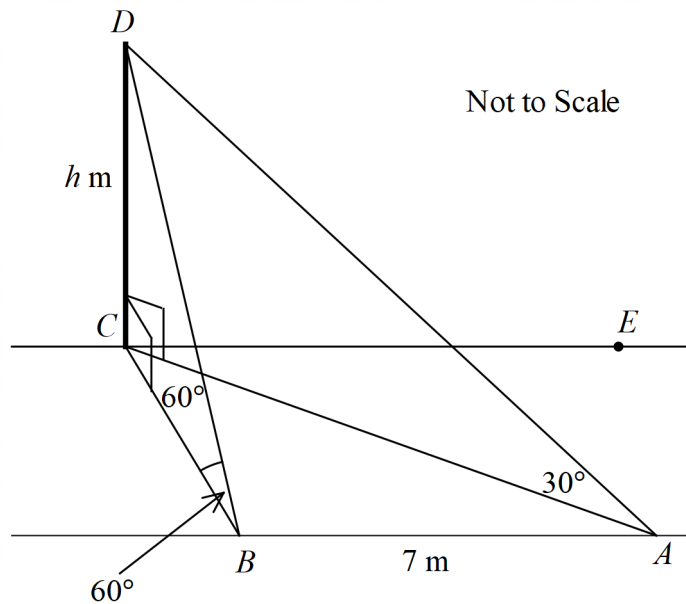
(b) Suppose $(7 + 3x)^{25} = \sum_{k=0}^{25} t_k x^k$.

- (i) Use the Binomial Theorem to write an expression for t_k , $0 \leq k \leq 25$. 1
- (ii) Show that $\frac{t_{k+1}}{t_k} = \frac{3(25-k)}{7(k+1)}$. 2
- (iii) Hence, or otherwise, find the largest coefficient t_k . You may leave your answer in the form $\binom{25}{k} 7^c 3^d$. 2

Question 13 continues on page 10

Question 13 (continued)

(c)



A footpath on horizontal ground has two parallel edges, AB and CE .

CD is a vertical flagpole of height h metres which stands with its base C on one edge of the footpath.

A and B are two points on the other edge of the footpath such that $AB = 7$ m and $\angle ACB = 60^\circ$. From A and B the angles of elevation of the top (D) of the flagpole are 30° and 60° respectively.

- | | | |
|------|--|---|
| (i) | Find the exact height of the flagpole. | 4 |
| (ii) | Find the width of the footpath correct to one decimal place. | 2 |

End of Question 13

- (a) Solve $\sqrt{3} \sin x - \cos x = 1$, $0 \leq x \leq 2\pi$, using $t = \tan \frac{x}{2}$ as a substitution. **3**

- (b) A cup of hot coffee at temperature $T^\circ\text{C}$ loses heat when placed in a cooler environment. It cools according to the law

$$T = T_0 + Ae^{-kt}$$

where t is time elapsed in minutes, and T_0 is the temperature of the environment in degrees in Celsius.

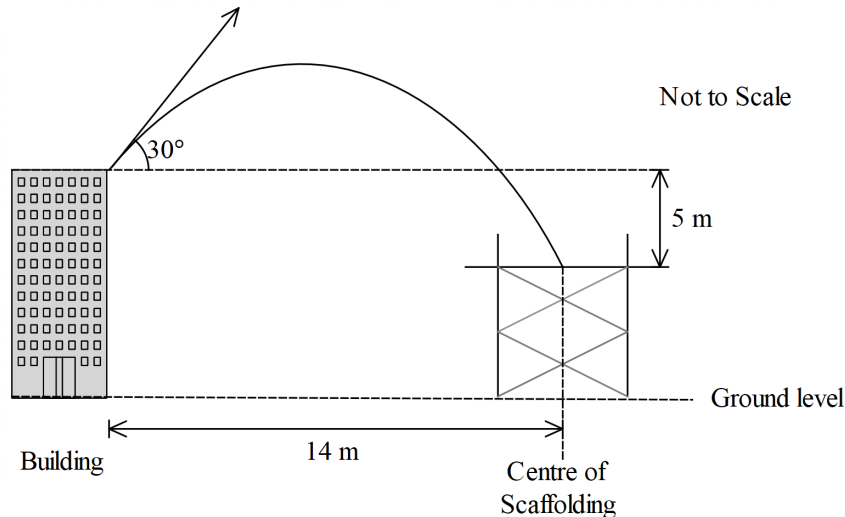
- (i) A cup of coffee at 100°C is placed in an environment at -20°C for 3 minutes, and cools to 70°C . Find the exact value of k . **2**
- (ii) The same cup of coffee, at 70°C , is then placed in an environment at 20°C . Assuming k remains the same, find the temperature of the coffee after a further 15 minutes. Give your answer to the nearest degree. **2**

- (c) Evaluate $\int_0^1 \frac{e^{\cos^{-1} x}}{\sqrt{1-x^2}} dx$ using the substitution $u = \cos^{-1} x$ **2**

Question 14 continues on page 12

Question 14 (continued)

- (d) A film director must decide whether a stuntman is able to perform a dangerous stunt. The stuntman must leap from a building onto the centre of some erected scaffolding. The centre of the scaffolding is 5 m below his initial position and at a horizontal distance of 14 m. The stuntman jumps at an angle of 30° above the horizontal. Let the stuntman's initial velocity be V , and let x and y be his horizontal and vertical displacements respectively from his initial position.



You may assume that the velocity and displacement equations are:

$$\begin{aligned}\dot{x} &= V \cos 30^\circ & \dot{y} &= -10t + V \sin 30^\circ \\ x &= Vt \cos 30^\circ & y &= -5t^2 + Vt \sin 30^\circ\end{aligned}$$

- (i) Show that the Cartesian equation of the stuntman's path is 2

$$y = \frac{-20x^2}{3V^2} + \frac{x}{\sqrt{3}}.$$

- (ii) Hence, determine the required initial velocity V , so that he lands in the centre of the scaffolding. Write your answer to the nearest m/s. 2

- (iii) Safety requirements are such that if the impact velocity is greater than 15m/s, then padding must be placed on the scaffolding. Assuming that the stuntman leaps at the required speed, determine whether padding is needed or not. 2

End of Paper

Extension 1 Trial 2019 Solⁿs and marking guidelines.

1. $A(2, -3)$ $B(-4, 5)$ $4:3$

$$\left(\frac{4x - 4 + 3 \times 2}{4 + 3}, \frac{4 \times 5 + 3x - 3}{4 + 3} \right)$$

$$= \left(\frac{-10}{7}, \frac{11}{7} \right) \quad \textcircled{A}$$

2. $P(x) = x^3 - 10x$

$$P(-5) = (-5)^3 - 10 \times -5$$

$$= -125 + 50$$

$$= -75$$

$\textcircled{A}$

3. $x = t - 3$

$$y = t^2 + 2$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= 2t \times 1$$

$$= 2t$$

When $t = -3$

$$x = -6$$

$$y = 11$$

$$\frac{dy}{dx} = -6$$

Eqⁿ of tangent

$$y - 11 = -6(x + 6)$$

$$6x + y + 25 = 0$$

$\textcircled{C}$

4. B

2019 Trial Exam Mathematics Ext 1. solutions.

Qu.

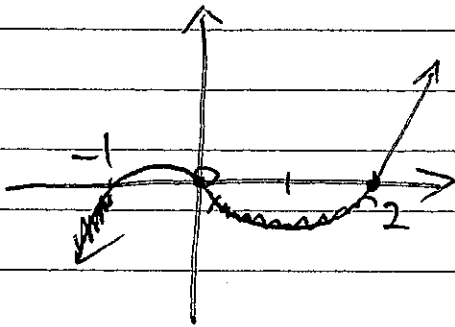
a). $x - \frac{2}{x} \leq 1, (x \neq 0)$ } multiply by x^2 .

$$x^3 - 2x \leq x^2$$

$$x^3 - x^2 - 2x \leq 0$$

$$x(x^2 - x - 2) \leq 0$$

$$x(x-2)(x+1) \leq 0$$



$$\therefore x \leq -1 \text{ or } 0 < x \leq 2$$

or.

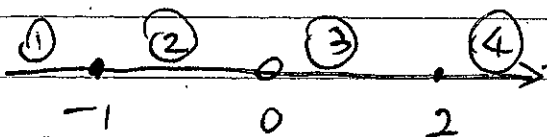
Consider $x - \frac{2}{x} = 1, x \neq 0$

$$x^2 - 2 = x$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

Critical values: $x = -1, 0, 2$.



Testing:

① when $x = -2$, $LHS = -2 + 1 = -1$ ✓

② when $x = -\frac{1}{2}$, $LHS = -\frac{1}{2} + 4 = 3\frac{1}{2}$ ✗

③ when $x = 1$, $LHS = 1 - 2 = -1$ ✓

④ when $x = 3$, $LHS = 3 - \frac{2}{3} = 2\frac{1}{3}$ ✗ $\therefore x \leq -1 \text{ or } 0 < x \leq 2$

1 mark.

Simplify with x^2 .

1 mark

Correct answer.

1 mark

Obtain the critical values

1 mark

Correct testing & answers.

⑪

$$b). \int_{-1}^0 x \sqrt{1+x} dx.$$

$$= \int_0^1 (u-1) \sqrt{u} du.$$

$$= \int_0^1 (u^{\frac{3}{2}} - u^{\frac{1}{2}}) du$$

$$= \left[u^{\frac{5}{2}} \div \frac{5}{2} - u^{\frac{3}{2}} \div \frac{3}{2} \right]_0^1$$

$$= \frac{2}{5} - \frac{2}{3}$$

$$= -\frac{4}{15}.$$

$$\text{let } u=1+x \\ du=dx.$$

$$\text{when } x=-1, u=0$$

$$\text{when } x=0, u=1.$$

$$x=u-1.$$

1 mark

correct substitution

1 mark.

correct integral

1 mark

correct answer.

$$c). \tan\left(\frac{5\pi}{12}\right)$$

$$= \tan\left(\frac{\pi}{6} + \frac{\pi}{4}\right)$$

$$= \frac{\tan\frac{\pi}{6} + \tan\frac{\pi}{4}}{1 - \tan\frac{\pi}{6}\tan\frac{\pi}{4}}$$

$$= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}}$$

$$= \frac{1 + \sqrt{3}}{\sqrt{3} - 1} \quad \swarrow \times \frac{\sqrt{3}}{\sqrt{3}}.$$

$$= \frac{(1 + \sqrt{3})(\sqrt{3} + 1)}{(\sqrt{3})^2 - 1}$$

$$= \frac{(\sqrt{3} + 1)^2}{2}$$

$$= \frac{4 + 2\sqrt{3}}{2}$$

$$= 2 + \sqrt{3}.$$

1 mark

correctly apply $\tan(\alpha+\beta)$

1 mark.

get the exact value

1 mark

simplify to the final answer.

d). i). In sector OAB: $\widehat{AB} = r\theta$.

$$AC = 2\widehat{AB} = 2r\theta.$$

$$\begin{aligned}\text{In } \triangle AOC, \tan \theta &= \frac{AC}{OA} \\ &= \frac{2r\theta}{r} \\ &= 2\theta.\end{aligned}$$

1 mark.
Use arc length: $l = r\theta$

1 mark
correct proof.

ii). Let $f(\theta) = \tan \theta - 2\theta$.

$$f'(\theta) = \sec^2 \theta - 2.$$

1 mark.
correct $f(\theta)$ & $f'(\theta)$

$$\theta_1 = \theta_0 - \frac{f(\theta_0)}{f'(\theta_0)}$$

1 mark

$$= 1.2 - \frac{\tan(1.2) - 2(1.2)}{\sec^2(1.2) - 2}$$

correct answer.

$$= 1.1693 \dots$$

$$\approx 1.17 \text{ (2dp)}.$$

2). Let $\angle QPR = \alpha$ & $\angle QRP = \beta$.

1 mark

$\angle QPR = \angle PBA = \alpha$ (angle between the tangent & chord equals to the angles at the circumference subtended by the alternate segment).

Apply Alternate Segment

Similarly $\angle QRP = \angle RBA = \beta$.

1 mark

Attempt to the final answer

$$\therefore \angle PBR = \alpha + \beta.$$

In $\triangle QPR$, $\angle PQR + \alpha + \beta = 180^\circ$ (angle sum of a triangle)

1 mark

Correct proof.

$$\therefore \angle PQR = 180 - (\alpha + \beta).$$

$$\begin{aligned} \angle PBR + \angle PQR &= 180 - (\alpha + \beta) + (\alpha + \beta) \\ &= 180. \end{aligned}$$

$\therefore PQRB$ is a cyclic quadrilateral.
(Opposite angles are supplementary).

Q12.

c). Let the statement be

$$(1 - \frac{2}{2 \times 3})(1 - \frac{2}{3 \times 4}) \dots (1 - \frac{2}{n(n+1)}) = (\frac{n+2}{3n}) \text{ for } n \geq 2, n \in \mathbb{Z}.$$

1 mark
Correctly prove it's true for $n=2$.

Prove the statement is true for $n=2$.

$$\begin{aligned} \text{LHS} &= (1 - \frac{2}{2 \times 3}) \\ &= 1 - \frac{1}{3} \\ &= \frac{2}{3}. \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \frac{2+2}{3(2)} \\ &= \frac{2}{3}. \end{aligned}$$

1 mark
Correct writing of assumption & the require to prove statement.

Assume the statement is true for $n=k, k \geq 2, k \in \mathbb{Z}$.

$$(1 - \frac{2}{2 \times 3})(1 - \frac{2}{3 \times 4}) \dots (1 - \frac{2}{k(k+1)}) = \frac{k+2}{3k}.$$

1 mark.
Correct proof.

To prove the statement is true for $n=k+1$.

$$(1 - \frac{2}{2 \times 3})(1 - \frac{2}{3 \times 4}) \dots (1 - \frac{2}{k(k+1)})(1 - \frac{2}{(k+1)(k+2)}) = \frac{k+3}{3(k+1)}.$$

$$\begin{aligned} \text{LHS} &= (1 - \frac{2}{2 \times 3})(1 - \frac{2}{3 \times 4}) \dots (1 - \frac{2}{k(k+1)})(1 - \frac{2}{(k+1)(k+2)}) \\ &= \frac{k+2}{3k} (1 - \frac{2}{(k+1)(k+2)}) \\ &= \frac{k+2}{3k} (\frac{(k+1)(k+2) - 2}{(k+1)(k+2)}) \\ &= \frac{k+2}{3k} (\frac{k^2 + 3k + 2 - 2}{(k+1)(k+2)}) \\ &= \frac{k+2}{3k} (\frac{k^2 + 3k}{(k+1)(k+2)}) \\ &= \frac{k+2}{3k} (\frac{k(k+3)}{(k+1)(k+2)}) \end{aligned}$$

$$= \frac{k+3}{3(k+1)} = \text{P.T.S.}$$

$\therefore$ The statement is true for $n=k+1$ if it is true for $n=k$.

Since the statement is true for $n=2$, it is true for $n=2+1=3$, $n=3+1=4$... By the concept of mathematical induction, it is true for all integers $n \geq 2$.

b) i). $m_{OP} = \frac{ap^2 - 0}{2ap} = \frac{p}{2}$ similarly $m_{OQ} = \frac{q}{2}$

Since $OP \perp OQ$, $m_{OP} \times m_{OQ} = -1$.

$$\frac{p}{2} \times \frac{q}{2} = -1.$$

$$pq = -4.$$

1 mark.

Correct proof.

ii). Since $OPRQ$ is a rectangle, the diagonals bisect each other.

$\therefore M$ is also the midpoint of OR .

1 mark

Correct proof.

$$\frac{0+x_R}{2} = a(p+q), \quad \frac{0+y_R}{2} = \frac{1}{2}a(p^2+q^2)$$

$$x_R = 2a(p+q).$$

$$y_R = a(p^2+q^2)$$

$$\therefore R(2a(p+q), a(p^2+q^2)).$$

$$\text{iii) } x = 2a(p+q).$$

$$\frac{x}{2a} = p+q.$$

$$\frac{x^2}{4a^2} = p^2 + q^2 + 2pq.$$

$$= p^2 + q^2 - 8 \quad \text{since } pq = -4.$$

$$y = a(p^2 + q^2)$$

$$\frac{y}{a} = p^2 + q^2.$$

$$\therefore \frac{x^2}{4a^2} = \frac{y}{a} - 8.$$

$$x^2 = 4a(y - 8a).$$

$$\text{c) i) } \frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}.$$

$$A = 4\pi r^2 \rightarrow \frac{dA}{dr} = 8\pi r, \quad \frac{dA}{dt} = 1.5$$

$$\therefore 1.5 = 8\pi r \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = 1.5 \times \frac{1}{8\pi r}$$

$$= \frac{3}{16\pi r} \text{ mm/s.}$$

$$\text{ii) } \frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}.$$

$$V = \frac{4}{3}\pi r^3 \rightarrow \frac{dV}{dr} = 4\pi r^2.$$

$$\frac{dV}{dt} = 4\pi r^2 \times \frac{3}{16\pi r}$$

$$= \frac{3r}{4}.$$

1 mark

Make $p+q$ or p^2+q^2

in terms of x and y .

1 mark

Correct equation.

1 mark.

Apply Chain Rule.

1 mark

Correct answer.

1 mark

Apply chain rule

1 mark

Correct answer.

When $r=60$, $\frac{dv}{dt} = 45 \text{ mm}^3/\text{s}$.

i). i). $v^2 = 36 - 4x^2$
 $\frac{1}{2}v^2 = 18 - 2x^2$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

$$= -4x.$$

$$= -2^2 x.$$

$\therefore$ It's simple harmonic motion since

$$\ddot{x} = -n^2 x.$$

1 mark
Correct answer.

ii). $\ddot{x} = -4x$

$$n^2 = 4$$

$$n = 2 \text{ since } n > 0.$$

$$\text{Period } T = \frac{2\pi}{n}$$

$$= \pi \text{ sec.}$$

1 mark
Get correct period
or amplitude.

For amplitude, make $v=0$.

$$36 - 4x^2 = 0.$$

$$x^2 = 9$$

$$x = \pm 3.$$

Since the centre of the motion is at $x=0$.

$\therefore$ Amplitude $a=3$.

1 mark
Correct answers

iii). $n=2$, $a=3$.

$$x = 3 \sin 2t \text{ or } x = -3 \sin 2t.$$

1 mark
Correct answer
(both positive & negative
are acceptable).

13 a) i) $v = -x^2$
 $v^2 = x^4$
 $\frac{1}{2} v^2 = \frac{1}{2} x^4$

2	correct sol ⁿ
1	one error in process

$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x^3$
 When $x = 1$ $\ddot{x} = 2 \times 1^3$
 $= \boxed{2 \text{ m/s}^2}$

ii) $\frac{dx}{dt} = -x^2$
 $\int -\frac{dx}{x^2} = \int dt$
 $\frac{1}{x} = t + C$

2	Correct sol ⁿ
1	one error in process

When $t = 0$ $x = 1$
 $\therefore C = 1$
 $\therefore \frac{1}{x} = t + 1$
 $\boxed{x = \frac{1}{t+1}}$

or $\frac{dx}{dt} = x^2$
 $\therefore \frac{dt}{dx} = -\frac{1}{x^2}$
 $\therefore t = \frac{1}{x} + C$
 $t = 0, x = 1 \therefore C = -1$
 $\therefore t = \frac{1}{x} - 1$
 $\therefore \frac{1}{x} = t + 1$
 $\therefore \boxed{x = \frac{1}{t+1}}$

b) i) $(7+3x)^{25} = 7^{25} + {}^{25}C_1 7^{24} (3x) + {}^{25}C_2 7^{23} (3x)^2 + \dots + (3x)^{25}$

$\therefore t_k = {}^{25}C_k \cdot 7^{25-k} \cdot 3^k$ or ${}^{25}C_k \cdot 7^{25-k} \cdot 3^k$

ii) $\frac{t_{k+1}}{t_k} = \frac{{}^{25}C_{k+1} \cdot 7^{25-(k+1)} \cdot 3^{k+1}}{{}^{25}C_k \cdot 7^{25-k} \cdot 3^k}$
 $= \frac{25!}{(25-(k+1))! (k+1)!} \times 7^{25-k-1-25+k} \times 3^{k+1-k}$
 $= \frac{25!}{(25-k)! k!}$

$$= \frac{25!}{(24-k)!(k+1)!} \times \frac{(25-k)! k!}{25!} \times \frac{1}{7} \times 3$$

$$= \frac{3(25-k)}{7(k+1)} \quad \text{As req'd}$$

2	sufficiently substantiated proof
1	partially substantiated proof

iii) when $\frac{t_{k+1}}{t_k} > 1$

$\frac{t_{k+1}}{t_k} < 1$ zero marks

$$\frac{3(25-k)}{7(k+1)} > 1$$

$$75 - 3k > 7k + 7 \quad k+1 > 0$$

$$-10k > -68$$

$$k < 6.8$$

$$k = 6$$

$$\therefore k = 6 \cdot T_{k+1} > T_k$$

$$\therefore T_7 = {}^{25}C_7 \cdot 7^{18} \cdot 3^7$$

2 correct solution

1 correct inequality and attempt to solve

c) i) In $\triangle ACD$ $\tan 30^\circ = \frac{h}{AC}$

In $\triangle BCD$

$$\tan 60^\circ = \frac{h}{BC}$$

$$AC = \frac{h}{\tan 30^\circ}$$

$$AC = \sqrt{3}h$$

$$BC = \frac{h}{\tan 60^\circ} = \frac{h}{\sqrt{3}}$$

In $\triangle ABC$

$$7^2 = BC^2 + AC^2 - 2 \times BC \times AC \cos 60^\circ$$

$$49 = \left(\frac{h}{\sqrt{3}}\right)^2 + (\sqrt{3}h)^2 - 2 \times \frac{h}{\sqrt{3}} \times \sqrt{3}h \times \frac{1}{2}$$

$$49 = \frac{h^2}{3} + 3h^2 - h^2$$

$$49 = \frac{7h^2}{3}$$

$$h = \sqrt{21} \text{ m}$$

$$\begin{aligned} 1 \quad AC &= \sqrt{3}h & \textcircled{a} \quad h &= \frac{AC}{\sqrt{3}} & \textcircled{a} \quad AC &= h \cot 30 \\ BC &= \frac{h}{\sqrt{3}} & h &= \sqrt{3}BC & BC &= h \cot 60 \end{aligned}$$

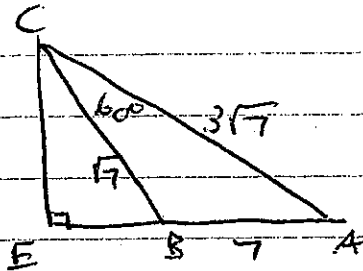
$$\textcircled{a} \quad \begin{aligned} AC &= \frac{h}{\tan 30} \\ BC &= \frac{h}{\tan 60} \end{aligned}$$

$$2 \quad \text{attempt sub in cos rule:} \\ 7^2 = BC^2 + AC^2 - 2BC \cdot AC \cdot \cos 60$$

$$3 \quad \begin{aligned} 49 &= \left(\frac{h}{\sqrt{3}}\right)^2 + (\sqrt{3}h)^2 - 2 \cdot \frac{h}{\sqrt{3}} \cdot \sqrt{3}h \cdot \frac{1}{2} \\ \cos 60 &= \frac{(\sqrt{3}h)^2 + \left(\frac{h}{\sqrt{3}}\right)^2 - 49}{2(\sqrt{3}h)\left(\frac{h}{\sqrt{3}}\right)} \end{aligned}$$

$$+ \text{ correct } h = \sqrt{21} \text{ m}$$

$$11) \quad h = \sqrt{21} \quad \therefore AC = \sqrt{3} \times \sqrt{21} \quad BC = \frac{\sqrt{21}}{\sqrt{3}} \\ = 3\sqrt{7} \quad = \sqrt{7}$$



In $\triangle ABC$

$$\begin{aligned} \frac{\sin A}{\sqrt{7}} &= \frac{\sin 60^\circ}{7} \\ \sin A &= \frac{\sqrt{21}}{14} \end{aligned}$$

$$A = 19.10660535^\circ$$

$$\angle CBE = 60 + 19.10660535^\circ$$

$$= 79.10660535^\circ$$

$$\sin(\angle CBE) = \frac{CE}{\sqrt{7}}$$

$$\begin{aligned} CE &= \sqrt{7} \sin(\angle CBE) \\ &= 2.598076211 \end{aligned}$$

$$\approx \boxed{2.6 \text{ m}}$$

$$\begin{aligned} 1 \quad CA &= 3\sqrt{7} \\ CB &= \sqrt{7} \\ &\text{or} \\ CA &= \sqrt{3} [\text{ANS}] \\ CB &= \frac{[\text{ANS}]}{\sqrt{3}} \text{ and} \\ &\text{uses sine rule} \end{aligned}$$

$$2 \quad \text{correct soln}$$

14 a) $\sqrt{3} \sin x - \cos x = 1$

$$\sqrt{3} \times \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} = 1$$

$$2\sqrt{3}t - 1 + t^2 = 1 + t^2$$

$$2\sqrt{3}t = 2$$

$$t = \frac{1}{\sqrt{3}}$$

$$\therefore \tan \frac{x}{2} = \frac{1}{\sqrt{3}}$$

$$\frac{x}{2} = \frac{\pi}{6}$$

$$x = \frac{\pi}{3}$$

1 correct sub into formula

$$\sqrt{3} \times \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} = 1$$

2 $x = \frac{\pi}{3}$ (ignored $x = \pi$)

3. showed $x = \pi$ subs. correctly to conclude

$$x = \frac{\pi}{3}, \pi$$

Check $x = \pi$ $\sqrt{3} \sin \pi - \cos \pi$
 $= 0 - (-1)$
 $= 1$

$\therefore x = \pi$ is also a solⁿ

$\therefore$ complete solⁿ

$$x = \frac{\pi}{3}, \pi$$

b) i) $\frac{dT}{dt} = k(T + 20)$

$t = 0$ $T = 100$, $T_0 = -20$

$$100 = -20 + Ae^0$$

$$A = 120$$

$$\therefore T = -20 + 120e^{-kt}$$

but $t = 3$, $T = 70$

$$70 = -20 + 120e^{-3k}$$

$$e^{-3k} = \frac{3}{4}$$

$$\therefore -3k = \log_e\left(\frac{3}{4}\right)$$

$$k = -\frac{1}{3} \log_e\left(\frac{3}{4}\right)$$

$$\therefore k = \frac{1}{3} \log_e \frac{4}{3}$$

$$1 \quad T = -20 + 120e^{-kt}$$

2 correct k

$T = 70 \quad T_0 = 20 \quad t = 0$

$\therefore 70 = 20 + Ae^0$

$A = 50$

$\therefore T = 20 + 50e^{-kt}$

at $k = \frac{1}{3} \log e^{3/4}$

$\therefore T = 20 + 50e^{(-\frac{1}{3} \log e^{3/4}) \times 15}$

$T = 31.86523435$

$T \doteq 32^\circ$

temperature $\doteq 32^\circ\text{C}$

1	$T = 20 + 50e^{-kt}$
2	correct soln after subst.

c) $\int_0^1 \frac{e^{\cos^{-1}x}}{\sqrt{1-x^2}} dx$

$u = \cos^{-1}x$
 $du = -\frac{1}{\sqrt{1-x^2}}$

$= \int_{\pi/2}^0 -e^u du$

when $x = 1 \quad u = \cos^{-1}1$

$= 0$
 $x = 0 \quad u = \cos^{-1}0$
 $= \pi/2$

$= \int_0^{\pi/2} e^u du$

$= \left[e^u \right]_0^{\pi/2}$

$= e^{\pi/2} - e^0$

$= e^{\pi/2} - 1$

2	correct solution
1	substitution of all elements

$$d) i) x = Vt \cos 30^\circ - (1) \quad y = -5t^2 + Vt \sin 30^\circ - (2)$$

$$\text{From (1)} \quad t = \frac{x}{V \cos 30^\circ}$$

$$= \frac{2x}{\sqrt{3}V} \quad \text{substitute into (2)}$$

$$y = -5 \left(\frac{2x}{\sqrt{3}V} \right)^2 + V \left(\frac{2x}{\sqrt{3}V} \right) \sin 30^\circ \quad \checkmark$$

$$= \frac{-20x^2}{3V^2} + \frac{2x}{\sqrt{3}} \times \frac{1}{2}$$

$$\boxed{y = \frac{-20x^2}{3V^2} + \frac{x}{\sqrt{3}}} \quad \text{As req'd } \checkmark$$

2	Correct Solution
1	Making t the subject and substituting with one error

$$ii) \text{ When } x = 14 \quad y = -5$$

$$-5 = \frac{-20 \times 14^2}{3V^2} + \frac{14}{\sqrt{3}} \quad \checkmark$$

$$-15\sqrt{3}V^2 = -3920\sqrt{3} + 42V^2$$

$$V^2(42 + 15\sqrt{3}) = 3920\sqrt{3}$$

$$V = \sqrt{\frac{3920\sqrt{3}}{42 + 15\sqrt{3}}}$$

$$= 9.993792608$$

$$\boxed{V \doteq 10 \text{ m/s}}$$

$$iii) \text{ Impact velocity} = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$$\text{from i) } t = \frac{2x}{\sqrt{3}V} \quad \text{if } V \doteq 10 \text{ from iii) then}$$

$$\text{When } x = 14 \quad t = \frac{28}{10\sqrt{3}}$$

$$= \frac{14}{5\sqrt{3}}$$

$$\dot{x} = 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3}$$

$$\dot{y} = \frac{-10 \times 14}{5\sqrt{3}} + 10 \times$$

$$= -\frac{28}{\sqrt{3}} + 5$$

$$v = \sqrt{(5\sqrt{3})^2 + \left(5 - \frac{28}{\sqrt{3}}\right)^2}$$

$$= \sqrt{199.675258}$$

$$\approx 14.13064959 \text{ m/s}$$

1 | attempts to use
 $t = \frac{28}{10\sqrt{3}}$

$$\dot{x} = 5\sqrt{3} \text{ and } y$$

$$\dot{y} = -\frac{28}{\sqrt{3}} + 5$$

and sub into

$$v^2 = \dot{x}^2 + \dot{y}^2$$

2 | correct soln

which is less than the threshold for padding requirement $\therefore$ no padding needed